

DDL and DML used in lab6

StarsDB

December 5, 2025

1 DDL:

```
1 CREATE DATABASE MNHS;
2 USE MNHS;
3
4 CREATE TABLE Patient (
5     IID INT PRIMARY KEY,
6     CIN VARCHAR(10) UNIQUE NOT NULL,
7     FullName VARCHAR(100) NOT NULL,
8     Birth DATE,
9     Sex ENUM('M', 'F') NOT NULL,
10    BloodGroup ENUM('A+', 'A-', 'B+', 'B-', 'O+', 'O-', 'AB+', 'AB-'),
11    Phone VARCHAR(15)
12 );
13
14 CREATE TABLE Hospital (
15     HID INT PRIMARY KEY,
16     Name VARCHAR(100) NOT NULL,
17     City VARCHAR(50) NOT NULL,
18     Region VARCHAR(50)
19 );
20
21 CREATE TABLE Department (
22     DEP_ID INT,
23     HID INT,
24     Name VARCHAR(100) NOT NULL,
25     Specialty VARCHAR(100),
26     PRIMARY KEY (DEP_ID),
27     FOREIGN KEY (HID) REFERENCES Hospital(HID)
28 );
29
30 CREATE TABLE Staff (
31     STAFF_ID INT PRIMARY KEY,
32     FullName VARCHAR(100) NOT NULL,
33     Status ENUM('Active', 'Retired') DEFAULT 'Active'
34 );
35
36 CREATE TABLE Workin (
37     STAFF_ID INT,
38     DEP_ID INT,
39     primary key(STAFF_ID, DEP_ID),
40     FOREIGN KEY (STAFF_ID) REFERENCES Staff(STAFF_ID),
41     FOREIGN KEY (DEP_ID) REFERENCES Department(DEP_ID)
42 );
43
44 CREATE TABLE ClinicalActivity (
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45     CAID INT PRIMARY KEY,
46     IID INT NOT NULL,
47     STAFF_ID INT NOT NULL,
48     DEP_ID INT NOT NULL,
49     Date DATE NOT NULL,
50     Time TIME,
51     FOREIGN KEY (IID) REFERENCES Patient(IID),
52     FOREIGN KEY (STAFF_ID) REFERENCES Staff(STAFF_ID),
53     FOREIGN KEY (DEP_ID) REFERENCES Department(DEP_ID)
54 );
55
56 CREATE TABLE Appointment (
57     CAID INT PRIMARY KEY,
58     Reason VARCHAR(100),
59     Status ENUM('Scheduled', 'Completed', 'Cancelled') DEFAULT 'Scheduled',
60     FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
61 );
62
63 CREATE TABLE Emergency (
64     CAID INT PRIMARY KEY,
65     TriageLevel INT CHECK (TriageLevel BETWEEN 1 AND 5),
66     Outcome ENUM('Discharged', 'Admitted', 'Transferred', 'Deceased'),
67     FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
68 );
69
70 CREATE TABLE Insurance (
71     INS_ID INT PRIMARY KEY,
72     Type ENUM('CNOPS', 'CNSS', 'RAMEd', 'Private', 'None') NOT NULL
73 );
74
75
76 CREATE TABLE Expense (
77     EXP_ID INT PRIMARY KEY,
78     INS_ID INT,
79     CAID INT UNIQUE NOT NULL,
80     Total DECIMAL(10, 2) NOT NULL CHECK (Total >= 0),
81     FOREIGN KEY (INS_ID) REFERENCES Insurance(INS_ID),
82     FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
83 );
84
85 CREATE TABLE Medication (
86     MID INT PRIMARY KEY,
87     Name VARCHAR(100) NOT NULL,
88     Form VARCHAR(50),
89     Strength VARCHAR(50),
90     ActiveIngredient VARCHAR(100),
91     TherapeuticClass VARCHAR(100),
92     Manufacturer VARCHAR(100)
93 );
94
95 CREATE TABLE Stock (
96     HID INT,
97     MID INT,
98     StockTimestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
99     UnitPrice DECIMAL(10, 2) CHECK (UnitPrice >= 0),
100    Qty INT DEFAULT 0 CHECK (Qty >= 0),
101    ReorderLevel INT DEFAULT 10 CHECK (ReorderLevel >= 0),
102    PRIMARY KEY (HID, MID, StockTimestamp),
103    FOREIGN KEY (HID) REFERENCES Hospital(HID),

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104     FOREIGN KEY (MID) REFERENCES Medication(MID)
105 );
106
107 CREATE TABLE Prescription (
108     PID INT PRIMARY KEY,
109     CAID INT UNIQUE NOT NULL,
110     DateIssued DATE NOT NULL,
111     FOREIGN KEY (CAID) REFERENCES ClinicalActivity(CAID)
112 );
113
114 -- Many to many relationship between prescription and medication entities
115 CREATE TABLE Includes (
116     PID INT,
117     MID INT,
118     Dosage VARCHAR(100),
119     Duration VARCHAR(50),
120     PRIMARY KEY (PID, MID),
121     FOREIGN KEY (PID) REFERENCES Prescription(PID),
122     FOREIGN KEY (MID) REFERENCES Medication(MID)
123 );
124
125 CREATE TABLE ContactLocation (
126     CLID INT PRIMARY KEY,
127     City VARCHAR(50),
128     Province VARCHAR(50),
129     Street VARCHAR(100),
130     Number VARCHAR(10),
131     PostalCode VARCHAR(10),
132     PhoneLocation VARCHAR(15)
133 );
134
135 -- Many to many relationship between Patient and Contact Location entities
136 CREATE TABLE have (
137     IID INT,
138     CLID INT,
139     PRIMARY KEY (IID, CLID),
140     FOREIGN KEY (IID) REFERENCES Patient(IID),
141     FOREIGN KEY (CLID) REFERENCES ContactLocation(CLID)
142 );

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2 DML:

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1  -- Altering patient table to add email attribute
2  ALTER TABLE Patient
3  ADD Email VARCHAR(100)
4  CHECK (Email LIKE '%@%._%');
5
6
7  -- Altering Staff table to add address column
8  ALTER TABLE Staff
9  ADD Address VARCHAR(50) ;
10 -- Inserting sample rows
11 -- Patient
12 INSERT INTO Patient (IID, CIN, FullName, Birth, Sex, BloodGroup, Phone, Email)
13 VALUES
14 (1, 'CA123456', 'Youssef El Idrissi', '1985-04-12', 'M', 'A+', '0612345678', 'y.
    elidrissi@casablanca.ma'),

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15 (2, 'RA789012', 'Fatima Zahra Bennani', '1990-09-25', 'F', 'O-', '0654321987', 'f.
    zahra@rabat.ma'),
16 (3, 'ME345678', 'Mohamed Amrani', '1978-01-30', 'M', 'B+', '0623456789', 'm.
    amrani@meknes.ma'),
17 (4, 'FE901234', 'Khadija El Fassi', '2000-06-15', 'F', 'AB+', '0678912345', 'k.
    elfassi@fes.ma'),
18 (5, 'MA567890', 'Rachid Toumi', '1995-03-10', 'M', 'O+', '0645789123', 'r.
    toumi@marrakech.ma');
19
20 -- Hospital
21 INSERT INTO Hospital (HID, Name, City, Region)
22 VALUES
23 (1, 'CHU Hassan II', 'Casablanca', 'Casablanca-Settat'),
24 (2, 'H pital Ibn Sina', 'Rabat', 'Rabat-Sal -K nitra'),
25 (3, 'H pital Mohammed V', 'Marrakech', 'Marrakesh-Safi'),
26 (4, 'H pital Regional Tanger', 'Tangier', 'Tanger-Tetouan-Al Hoceima'),
27 (5, 'Centre Hospitalier Universitaire F s', 'F s', 'F s-Mekn s');
28
29 -- Department
30 INSERT INTO Department (DEP_ID, HID, Name, Specialty) VALUES
31 (1, 1, 'Cardiology', 'Heart diseases'),
32 (2, 2, 'Neurology', 'Brain and nerves'),
33 (3, 3, 'Oncology', 'Cancer treatment'),
34 (4, 4, 'Pediatrics', 'Child healthcare'),
35 (5, 5, 'Orthopedics', 'Bone and muscles');
36
37
38 -- Staff
39 INSERT INTO Staff (STAFF_ID, FullName, Status, Address) VALUES
40 (1, 'Dr. Salma El Alami', 'Active', '12 Boulevard Mohammed V, Casablanca'),
41 (2, 'Dr. Yassine Najjar', 'Active', '45 Rue Al Qods, Rabat'),
42 (3, 'Dr. Amina Boulahlib', 'Active', '3 Jamaa el Fna Sq, Marrakech'),
43 (4, 'Dr. Rami Sedrati', 'Retired', '8 Medina St, Tangier'),
44 (5, 'Dr. Laila Chraibi', 'Active', '20 Andalus Ave, F s');
45
46 -- Workin
47 INSERT INTO Workin (STAFF_ID, DEP_ID)
48 VALUES
49 (1, 1),
50 (2, 2),
51 (3, 3),
52 (4, 4),
53 (5, 5);
54
55 -- ClinicalActivity
56 INSERT INTO ClinicalActivity (CAID, IID, STAFF_ID, DEP_ID, Date, Time)
57 VALUES
58 (1, 1, 1, 1, '2025-11-10', '09:30:00'),
59 (2, 2, 2, 2, '2025-11-11', '14:00:00'),
60 (3, 3, 3, 3, '2025-11-12', '10:15:00'),
61 (4, 4, 4, 4, '2025-11-13', '16:45:00'),
62 (5, 5, 5, 5, '2025-11-14', '08:00:00');
63
64 -- Appointment
65 INSERT INTO Appointment (CAID, Reason, Status)
66 VALUES
67 (1, 'Routine checkup', 'Completed'),
68 (2, 'Headache evaluation', 'Scheduled'),
69 (3, 'Cancer screening', 'Completed'),

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70 (4, 'Vaccination', 'Cancelled'),
71 (5, 'Bone fracture follow-up', 'Completed');
72
73 -- Emergency
74 INSERT INTO Emergency (CAID, TriageLevel, Outcome)
75 VALUES
76 (1, 3, 'Discharged'),
77 (2, 2, 'Admitted'),
78 (3, 4, 'Transferred'),
79 (4, 1, 'Deceased'),
80 (5, 5, 'Discharged');
81
82 -- Insurance
83 INSERT INTO Insurance (INS_ID, Type)
84 VALUES
85 (1, 'CNOPS'),
86 (2, 'CNSS'),
87 (3, 'RAMED'),
88 (4, 'Private'),
89 (5, 'None');
90
91 -- Expense
92 INSERT INTO Expense (EXP_ID, INS_ID, CAID, Total)
93 VALUES
94 (1, 1, 1, 1200.50),
95 (2, 2, 2, 800.00),
96 (3, 3, 3, 1500.75),
97 (4, 4, 4, 500.25),
98 (5, 5, 5, 300.00);
99
100 -- Medication
101 INSERT INTO Medication (MID, Name, Form, Strength, ActiveIngredient,
102     TherapeuticClass, Manufacturer)
103 VALUES
104 (1, 'Paracetamol', 'Tablet', '500mg', 'Paracetamol', 'Analgesic', 'Pharma Maroc'),
105 (2, 'Amoxicillin', 'Capsule', '250mg', 'Amoxicillin', 'Antibiotic', 'Sandoz Maroc'),
106 (3, 'Ibuprofen', 'Tablet', '400mg', 'Ibuprofen', 'Anti-inflammatory', 'Rameda'),
107 (4, 'Omeprazole', 'Capsule', '20mg', 'Omeprazole', 'Proton pump inhibitor', 'Medpharm Maroc'),
108 (5, 'Cetirizine', 'Tablet', '10mg', 'Cetirizine', 'Antihistamine', 'Pharma Plus');
109
110
111 -- Stock
112 INSERT INTO Stock (HID, MID, StockTimestamp, UnitPrice, Qty, ReorderLevel)
113 VALUES
114 (1, 1, '2025-11-01 08:00:00', 0.5, 100, 20),
115 (2, 2, '2025-11-02 09:00:00', 1.2, 50, 10),
116 (3, 3, '2025-11-03 10:00:00', 0.8, 75, 15),
117 (4, 4, '2025-11-04 11:00:00', 1.5, 30, 5),
118 (5, 5, '2025-11-05 12:00:00', 0.7, 90, 25);
119
120 -- Prescription
121 INSERT INTO Prescription (PID, CAID, DateIssued)
122 VALUES
123 (1, 1, '2025-11-10'),
124 (2, 2, '2025-11-11'),
125 (3, 3, '2025-11-12'),

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126 (4, 4, '2025-11-13'),
127 (5, 5, '2025-11-14');
128
129 -- Includes
130 INSERT INTO Includes (PID, MID, Dosage, Duration)
131 VALUES
132 (1, 1, '500mg twice a day', '5 days'),
133 (2, 2, '250mg three times a day', '7 days'),
134 (3, 3, '400mg once a day', '3 days'),
135 (4, 4, '20mg once a day', '14 days'),
136 (5, 5, '10mg once a day', '10 days');
137
138 -- ContactLocation
139 INSERT INTO ContactLocation (CLID, City, Province, Street, Number, PostalCode,
140 PhoneLocation) VALUES
141 (1, 'Casablanca', 'Casablanca-Settat', 'Mohammed V Blvd', '12', '20000', '
142 0612345678'),
143 (2, 'Rabat', 'Rabat-Sal -K nitra', 'Al Qods St', '45', '10000', '0654321987'),
144 (3, 'Marrakech', 'Marrakesh-Safi', 'Jamaa el Fna Sq', '3', '40000', '0623456789'),
145 (4, 'Tangier', 'Tanger-Tetouan-Al Hoceima', 'Medina St', '8', '90000', '0678912345
146 '),
147 (5, 'F s', 'F s-Mekn s', 'Andalus Ave', '20', '30000', '0645789123');
148
149 -- have
150 INSERT INTO have (IID, CLID)
151 VALUES
152 (1, 1),
153 (2, 2),
154 (3, 3),
155 (4, 4),
156 (5, 5);

```